



DSA 3050A US 2026 – BUSINESS INTELLIGENCE & DATA VISUALIZATION

END SEMESTER PRACTICAL EXAMINATION

Duration: 72 Hours

Total Marks: 100 Marks

Software: Microsoft Power BI Desktop

Submission: GitHub Repository

Examination Brief

You are required to design and develop a complete **Business Intelligence solution using Microsoft Power BI**.

You will identify and obtain your **own real-world dataset**, prepare and transform the data using Power Query, develop an appropriate data model, create DAX calculations, and build professional interactive dashboards that communicate meaningful insights.

This examination assesses your ability to move from **raw data to business intelligence**, rather than simply producing visualizations.

Your final work must demonstrate:

- Data acquisition and understanding
- Data cleaning and transformation using Power Query
- Data modelling and relationships
- DAX calculations and analytical measures
- Dashboard design and storytelling
- Business interpretation
- GitHub documentation and evidence

SECTION A: DATASET SELECTION & UNDERSTANDING – 10 MARKS

Select a real-world dataset from a credible source such as Kaggle, government open-data portals, organizational open data, World Bank, WHO or another verifiable source.

The dataset should be sufficiently complex to support meaningful BI analysis.

Minimum requirements

Your dataset should preferably contain:

- **At least 20,000 records**
- Multiple numerical variables
- Multiple categorical variables
- Date/time information
- At least one variable that can be used to calculate a KPI
- Enough complexity to support multiple dimensions of analysis

Datasets that are already completely cleaned and summarized should be avoided.

In your README .md, explain:

1. The source of your dataset.
2. What the dataset represents.
3. Why you selected it.
4. The main variables available.
5. The business/analytical problem you intend to investigate.
6. At least **five analytical questions** that your Power BI solution should answer.

Marks: 10

SECTION B: POWER QUERY – DATA CLEANING & TRANSFORMATION – 20 MARKS

Use **Power Query extensively** to transform the raw data into a form suitable for analysis.

Simply importing a clean CSV/Excel file and clicking **Close & Apply** will receive very limited marks.

Your transformations should be based on the problems found in **your own dataset**.

You may consider operations such as:

- Removing unnecessary columns
- Handling missing/null values
- Removing duplicates
- Correcting data types
- Standardizing inconsistent categories
- Cleaning text
- Replacing incorrect values
- Splitting columns

- Merging columns
- Creating conditional columns
- Creating custom columns
- Extracting information from dates
- Grouping records
- Pivoting/unpivoting
- Merging queries
- Appending queries
- Creating reference/duplicate queries
- Creating dimension tables
- Handling errors
- Renaming fields appropriately

You are **not expected to perform every operation above**. Select transformations that are appropriate to your dataset.

Documentation requirement

For **at least 8 significant Power Query transformations**, explain:

Problem → Transformation → Reason → Result

For example:

Problem: The `Customer_Type` field contained values such as "Corporate", "CORPORATE" and "corporate".

Transformation: Used Transform → Format and Replace Values to standardize the categories.

Reason: These values represent the same customer category and would otherwise appear as separate groups during analysis.

Result: The field now contains one consistent "Corporate" category.

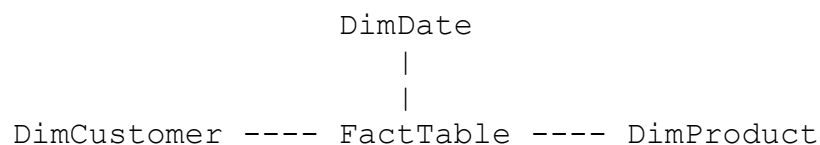
Include screenshots showing important stages of your Power Query work.

SECTION C: DATA MODELLING – 20 MARKS

Transform your cleaned data into an appropriate analytical model.

Your model should **not simply consist of one large flat table** unless you provide a convincing technical justification.

Where appropriate, develop a **Star Schema** consisting of:



|
|
DimLocation

The actual dimensions will depend on your dataset.

Requirements

You should demonstrate:

1. Identification of the main **fact table**.
2. Creation of appropriate **dimension tables**.
3. Primary/key fields suitable for relationships.
4. Appropriate table relationships.
5. Correct relationship cardinality.
6. Appropriate cross-filter direction.
7. A dedicated **Date Table** where the dataset supports time analysis.
8. Correct use of the Date Table in the model.
9. Appropriate data types.
10. Clear table and field naming.

Your model should avoid unnecessary **many-to-many relationships**, ambiguous filter paths and inappropriate bidirectional relationships.

Model Explanation

In your README, explain your model rather than merely showing a screenshot.

For example:

FactSales contains transactional information and forms the centre of the model.

DimProduct, DimCustomer, DimLocation and DimDate provide descriptive attributes used to filter the fact table. One-to-many relationships were established between each dimension and the fact table.

You must also explain:

- Why you selected your fact table.
- Why each dimension was created.
- The relationships used.
- Cardinality decisions.
- Filter direction decisions.
- Any modelling challenges encountered.

A clear screenshot of the completed **Model View** must be included.

SECTION D: DAX & BUSINESS CALCULATIONS – 20 MARKS

Create DAX calculations that turn your data model into a useful analytical solution.

You are required to create a minimum of **12 meaningful DAX measures**.

The measures must be relevant to **your dataset**. Therefore, students are **not expected to have identical measures**.

Level 1 – Core Measures

Create measures for fundamental KPIs relevant to your dataset.

Examples include:

```
Total Sales = SUM(FactSales[SalesAmount])
Total Transactions = COUNTROWS(FactSales)
Average Sales = AVERAGE(FactSales[SalesAmount])
```

Level 2 – Calculated Business Measures

Develop measures that derive additional information.

Examples:

```
Total Profit =
[Total Sales] - [Total Cost]
Profit Margin % =
DIVIDE(
    [Total Profit],
    [Total Sales],
    0
)
```

Students should select calculations appropriate to their own business problem.

Level 3 – Advanced DAX

Your solution must contain evidence of more advanced DAX.

Depending on your dataset, consider:

- CALCULATE()
- FILTER()
- DIVIDE()
- ALL()

- ALLSELECTED()
- DISTINCTCOUNT()
- SWITCH()
- IF()
- Variables using VAR
- Ranking using RANKX()
- Time Intelligence
- Context manipulation

For datasets containing suitable dates, you could calculate:

```
Previous Year Sales =
CALCULATE (
    [Total Sales],
    SAMEPERIODLASTYEAR (DimDate[Date])
)
```

and:

```
YoY Growth % =
DIVIDE (
    [Total Sales] - [Previous Year Sales],
    [Previous Year Sales],
    0
)
```

These examples are provided only to demonstrate the **expected level of complexity**. Students must develop calculations appropriate to their own dataset.

DAX Documentation

Select your **six most important DAX measures** and explain:

1. What the measure calculates.
2. Why the measure is useful.
3. The main DAX functions used.
4. How filter context affects the result.
5. Where the measure is used in your dashboard.

Do not simply paste DAX code without explaining its purpose.

SECTION E: PROFESSIONAL POWER BI DASHBOARDS – 20 MARKS

Develop a minimum of **three well-designed report pages**.

You are free to determine the purpose of each page based on your dataset.

However, your report should generally move from:

Overview → Detailed Analysis → Deeper Insights

Page 1: Executive Overview

This should provide management with an immediate understanding of performance.

Consider:

- KPI cards
- Major trends
- Overall performance
- Important categories
- Geographic performance
- Key comparisons
- Appropriate slicers

A manager should be able to look at this page and understand the major story within a few seconds.

Page 2: Detailed Analysis

Select an important area of your dataset for deeper analysis.

Examples:

- Product Analysis
- Customer Analysis
- Financial Analysis
- Regional Analysis
- Employee Analysis
- Healthcare Analysis
- Education Analysis
- Transport Analysis

Page 3: Advanced/Diagnostic Analysis

Investigate **why something is happening**, rather than simply showing what happened.

This page should demonstrate deeper analytical thinking.

Dashboard Quality Expectations

Your dashboards will be assessed on more than the number of charts included.

Visual selection

Choose visuals appropriate to the question being answered.

Students should avoid using inappropriate visuals simply to demonstrate that they know how to create them.

Layout

Your dashboard should demonstrate:

- Alignment
- Consistent spacing
- Clear visual hierarchy
- Appropriate use of screen space
- Readable fonts
- Consistent formatting

Colour

Use a professional and consistent colour palette.

Avoid excessive colours that do not communicate meaning.

Interactivity

Your report should demonstrate meaningful use of Power BI interactivity such as:

- Slicers
- Cross-filtering
- Drill-down
- Drill-through
- Report-page tooltips
- Bookmarks
- Buttons/navigation
- Dynamic titles

You do **not** need to use all these features, but the dashboard should demonstrate more than basic static charts.

Storytelling

Your pages should collectively tell a story.

A good dashboard should allow the user to move from:

What happened? → Where did it happen? → Why did it happen? → What requires attention?

Dashboard Assessment

SECTION F: GITHUB, SCREENSHOTS & DOCUMENTATION – 10 MARKS

Create a GitHub repository named using the following convention:

DSA3050-PowerBI-StudentName-RegNo

Example:

DSA3050-PowerBI-Austin-123456

Recommended structure:

```
DSA3050-PowerBI-StudentName-RegNo/  
├── README.md  
├── data/  
│   └── dataset.csv  
├── powerbi/  
│   └── DSA3050_StudentName.pbix  
└── screenshots/  
    ├── 01_raw_data.png  
    ├── 02_power_query.png  
    ├── 03_model.png  
    ├── 04_dashboard_overview.png  
    ├── 05_dashboard_analysis.png  
    └── 06_dashboard_insights.png
```

The README should document the entire BI development process:

Dataset → Power Query → Model → DAX → Dashboard → Insights

Screenshots must be clear and readable. Students should include screenshots of:

- Dataset/raw data
- Power Query transformations
- Completed data model
- DAX measures
- Each dashboard page
- Important analytical results

The repository must contain the .pbix file, dataset/source information, screenshots and README.md.

Students should demonstrate **progressive development through meaningful Git commits**, rather than uploading the complete project in one final commit.

Important Examination Conditions

This is an **individual practical examination**.

Students may use online documentation and learning resources, but the submitted Power BI solution must demonstrate their own understanding and analytical decisions.

Students using the same dataset may still have different transformations, models, measures and dashboards. **Identical Power BI files, identical DAX implementations, identical dashboard layouts or substantially identical GitHub repositories will be investigated.**

Students should be prepared to explain any Power Query transformation, modelling decision, DAX measure or dashboard feature submitted as part of their work.

The **72-hour period is intended for analysis, development, refinement and documentation**. Marks will therefore favour well-justified technical decisions and polished dashboards rather than simply the number of transformations, DAX formulas or visuals produced.